Toheeb Goodluck Ogunade

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Education

University of Lagos Lagos, Nigeria
B.Sc. Computer Science | *CGPA: 4.62/5.00 (First Class — top of a 5-point scale)* Expected Sep 2028

- **Relevant Coursework:** Data Structures & Algorithms, Concurrent Programming, Distributed Systems, Database Architecture, Systems Programming.

Skills

Languages:	Go, Python, SQL/PostgreSQL, TypeScript, C++
Databases & Stores:	PostgreSQL, BadgerDB, Redis, MongoDB
Tools & Infra:	Docker, Git, Linux, CI/CD, Kubernetes, OpenTelemetry (OTel), Swagger Docs
Frameworks:	NestJS, Node.js, FastAPI, Go Fiber, Express, React, Django

Experience

Software Engineering Intern *(Fraud Monitoring Team)* Jun 2026 – Dec 2026
Moniepoint, *Lagos, Nigeria*

- Engineered and integrated an **OpenTelemetry (OTel)** observability pipeline using **NestJS** and **Node.js**, deploying over 1,400 lines of **infrastructure code** to enable **full-stack traceability** for core operations.
- Developed a **centralized observability service** that processed 500+ unique telemetry events per second, directly feeding compressed data into **New Relic** for immediate anomaly detection.
- Implemented a **global HTTP trace interceptor** and **console logger**, embedding **trace IDs** directly into every log line to streamline **cross-service debugging** and reduce root-cause analysis bottlenecks by **40%**.
- Orchestrated secure cross-environment deployments across **3 active environments** by managing **Docker ConfigMaps** and **environment variables**, accelerating development speed and ensuring zero disruption to existing **CI/CD pipelines**.

Full-Stack Engineering Training May 2025 – Dec 2025
NITDA ICT Hub, *Lagos, Nigeria*

- Orchestrated the development and deployment of concurrent **microservices** using **Docker** and **Node.js**, streamlining system integrations and reducing overall deployment bottlenecks by **50%**.
- Built a **data aggregation system** with **normalized PostgreSQL schemas**, optimizing inefficient queries to increase **read throughput by 70%** across distributed, multi-tenant application layers.

Projects

Velarix — Decision-Integrity Engine | *Go, Python, TypeScript* https://github.com/qeinstein/velarix

- Created an **epistemic reasoning engine** and **symbolic interpretability layer**, providing 100% verifiable causal traces of stateful logic via an **OR-of-AND justification graph**.
- Engineered a concurrent **fact-extraction sidecar** achieving a **p50 latency of 35.6 ms** and a **p99 latency of 56.3 ms** with **96.3% triple recall**, optimizing throughput without relying on costly **LLM inference**.
- Built a secure, **PostgreSQL-backed HTTP API** utilizing **OAuth 2.0** and **JWT authentication**, processing up to **200 concurrent requests** and shipping integration surfaces for **LangGraph** alongside a specialized **Python SDK**.

TurboQuant — Implementation of Google Research's KV Cache Quantization Method (arXiv:2504.19874) | *Python, PyTorch*
https://github.com/qeinstein/TurboQuant

- Implemented the **TurboQuant** algorithm (arXiv:2504.19874) with a **QJL**-based inner-product corrector (arXiv:2406.03482): a two-stage **KV cache quantization** pipeline combining **Lloyd-Max scalar quantization** with a 1-bit **QJL** corrector, achieving **4.7$\times$ compression** at 4-bit keys + 2-bit values with cosine similarity $\sim$0.92 vs **fp16**.
- Identified and empirically confirmed an original finding absent from all three source papers: doubling the **QJL sketch size** to $m = 2d$ cuts the **perplexity gap** from $+21.11$ to $+9.83$ at 4-bit keys — less than half the quality loss, at the cost of a 2$\times$ larger sketch matrix.

Achievements & Extracurricular Activities

- **Technical Instructor (Volunteer):** StackD & University of Lagos ECX (Jan 2026 – Present). Taught Data Structures & Algorithms and backend engineering to over 50 computer science students by designing and delivering a 12-week technical curriculum.
- **Winner, Best AI Innovation – Curacel Hackathon (2024):** Pioneered the development of an intelligent bot leveraging customer interaction data to provide personalized responses, achieving a 20% improvement in first-contact resolution rates.
- **8th Place Nationally – National Mathematics Olympiad (2023):** Ranked 8th nationally by solving complex combinatorics, advanced algebra, and algorithmic optimization problems under strict time constraints.